Jose Manuel Martínez Morales

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Summary

Full-stack Software Engineer with a B.S. in Computer Science from Tecnológico de Monterrey who takes products from a blank page to production in small teams. Comfortable across the whole stack: Python, Java, and PHP backends, React frontends, PostgreSQL data modeling, automation pipelines, and Docker and cloud deployment. Turns manual business processes into reliable software, works closely with non-technical teams, and learns fast, promoted from intern to developer in three months. Native Spanish and professional English speaker with hands-on LLM research experience.

Education

Tecnológico de Monterrey — Guadalajara, Mexico
B.S. in Computer Science and Technology, GPA: 3.6/4.0 — *Aug. 2022 – Jun. 2026*
- Relevant Coursework: Data Structures & Algorithms, Distributed Systems, Databases, Cloud Computing, Machine Learning

Beijing Institute of Technology, top-tier Project 985 engineering university — Beijing, China
Competitively Selected International Exchange, Computer Science — *Sep. 2025 – Jan. 2026*

Experience

EVA — Remote
Software Engineer, Contract: Colosso Energy Platform — *Jul. 2026 – Present*
- Built Colosso Energy, a full-stack platform that turns per-circuit eGauge meter data into monthly energy reports and CFE-style tenant bills for about 15 EVA clients, replacing manual Excel reporting.
- Built a Python robot that polls every meter each 15 minutes and stores hourly kWh in a month-partitioned PostgreSQL table, tagged by CFE peak, intermediate, and base periods.
- Implemented CFE GDMTH and GDMTO tariff rules with exact Decimal math and generated PDF reports and tenant bills with ReportLab, emailed automatically each month.
- Wrote a CFE portal scraper with curl_cffi and BeautifulSoup that solves captchas with ddddocr and reads login codes over IMAP, then cross-checks real bills against meter data.
- Developed a Flask REST API and React 19 portal with 4 user roles enforced per request and through Supabase RLS, shipped as 3 Docker Compose containers behind Caddy.

FLI Holding — Guadalajara, Mexico
Software Developer, promoted from Software Development Intern in Dec. 2024 — *Sep. 2024 – May 2025*
- Promoted to developer in three months and became the first software hire, helping launch the in-house IT area.
- Co-built Humara, a BambooHR-style HR SaaS in PHP/Laravel used by 50+ employees and packaged for sale to external companies, as one of two engineers.
- Built full-stack modules with Blade, Tailwind CSS, and Laravel: time-off, rewards, profiles, and multi-level approvals.
- Shipped the full backlog in five months of two-week sprints with code reviews, deploying on Google Cloud.
- Built Looker Studio KPI dashboards and n8n/Zapier automations that send Humara alerts to Slack.

Projects

KernelForge, LLM Research with C3.ai | *Python, vLLM, XGrammar, PyTorch, Triton* — Mar. 2026 – Jun. 2026
- Co-authored a C3.ai paper showing grammar-constrained decoding improves the correctness of LLM-generated GPU kernels.
- Built a pipeline that generates and runs Triton kernels from GPT-5.4, DeepSeek V4 Pro, and Gemma 4 E4B using XGrammar.
- Created KAGBench, a 166-kernel benchmark with hidden Pytest tests, to compare each model with and without constraints.

Oracle Task Management System | *React, Java, Spring Boot, Kubernetes, OCI* — Feb. 2025 – May 2025
- Led the architecture of a microservices task platform for Oracle: React frontend and Spring Boot services on OCI Kubernetes.
- Built manager and employee views with team KPIs, a Telegram bot, JUnit tests, and CI/CD in GitHub Actions.

Awards

GuadalaHacks 2025, 1st Place of 30+ Teams | *Python, TensorFlow, HERE API, React* — Jan. 2025
- Led a team of 4 building a CNN that flags misplaced HERE map points of interest in satellite images, replacing manual QA.

Technical Skills

Languages: Python, Java, JavaScript, TypeScript, PHP, SQL, C++
Web & Backend: React, Flask, FastAPI, Spring Boot, Laravel, Node.js, Angular, Tailwind CSS, REST APIs
Data & Automation: PostgreSQL, Supabase, MySQL, MongoDB, ETL, Web Scraping, Kafka, n8n, Looker Studio
Cloud & DevOps: Docker, Docker Compose, Google Cloud, OCI, AWS, Kubernetes, Linux, Git, GitHub Actions
AI/ML: LLM APIs, Constrained Decoding, vLLM, PyTorch, Triton, TensorFlow, Computer Vision
Testing: Pytest, Jest, JUnit | **Spoken**: Spanish (native), English (professional), Mandarin (HSK 2)

Certifications

Oracle Cloud Infrastructure (OCI) Foundations Associate, Jun. 2025 | AWS Academy Cloud Foundations, May 2026